

Verification of the numerical methods: SRA1 vs ode45, with and without noise, and Benettin vs QR

Spiking-rate network with SFA and STD, preset
celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond_mu8p25

2026-09-11

Executive summary

Run. `numerics_verification_trials_run`, 2026-09-11, 51 min on 13 workers. Five network seeds for the integrator-precision experiments, twenty-five for the two Lyapunov comparisons, in each of the preset’s three adaptation regimes: no adaptation (strongly chaotic, LLE about +3.6), single-timescale adaptation (weakly chaotic and intermittent, LLE about +0.6), and multiple-timescale adaptation (stable, LLE about -0.12).

SRA1 vs ode45 without noise. At the paper’s 400 Hz, SRA1 reset every 0.02 s to a $1e-10$ ode45 reference accumulates a per-state error of $3.8e-5$ (chaotic), $2.6e-6$ (single-timescale) and $7.3e-8$ (multiple-timescale), i.e. relative errors of $1e-5$, $5e-6$ and $3e-7$. On every one of the 15 seed x regime combinations the error falls with step size at slope 2.00-2.01, the exact order of SRA1’s drift. The piecewise activation costs nothing.

SRA1 with the paper’s noise. On a shared Brownian path, SRA1 at 400 Hz differs from SRA1 at 12800 Hz by $4.1e-5$, $1.2e-5$ and $4.8e-6$ per state in the three regimes, converging at slopes 1.98-2.00, 1.92-1.97 and 1.82-1.88. The drift error dominates at $\sigma_u = 0.025$; the stochastic term is just visible where the drift error is smallest. The integrator behaves the same with the noise on as off.

Does the LLE depend on the integrator? No. Paired over 25 seeds, Benettin with SRA1 minus Benettin with ode45 is -0.03 ± 0.45 (chaotic), $+0.05 \pm 0.17$ (single-timescale) and 0.0000 ± 0.0000 (stable). Standard errors of those means are 0.09, 0.03 and 0.00; the signs split 14/11, 15/10 and 15/10. In the two chaotic regimes the per-seed scatter is large, because after a few Lyapunov times the two integrators follow different trajectories on the same attractor and a 10 s finite-time exponent depends on the trajectory; the ensemble means agree to 0.03 and 0.05, and the seed-to-seed spread (sd 0.62 and 0.17) is what the paper’s reps average over. In the stable regime the two integrators give the same exponent to four decimal places on every seed.

Benettin vs QR. On a 40-neuron network with the preset’s physics, paired over 25 seeds, QR’s largest exponent minus Benettin’s LLE is $+0.010 \pm 0.065$ (median |difference| 0.005; the exponents span -10 to +2 and correlate at 0.9999), $+0.015 \pm 0.058$ (median 0.031; span -0.65 to +0.64; correlation 0.985) and $+0.0001 \pm 0.020$ (median 0.005). The two independent estimators agree across a hundred-fold range of exponents. The one systematic feature is in the stable regime: QR’s λ_1 is -0.117 ± 0.004 on every seed (the slowest adaptation recovery), while Benettin scatters ± 0.02 with two low outliers, because Benettin’s perturbation has not always aligned with the slowest direction within the 10 s warm-up. That is a known finite-time property of Benettin’s method in strongly stable systems, and it biases the estimate toward more negative values, never toward instability.

Bottom line for the manuscript. The 400 Hz SRA1 integrator is verified in every regime with and without noise; its Lyapunov exponent is integrator-independent in the ensemble mean; and the Benettin estimate is corroborated by the full-spectrum QR method. Single-realisation exponents in the chaotic regimes carry finite-time scatter (sd 0.6 and 0.17 over 10 s), which is a property of the dynamics, not of the numerics; report medians over reps, as the sweeps do.

1. What was verified, and why

The manuscript’s simulations use a fixed-step stochastic integrator (Roessler SRA1 at 400 Hz) on a 500-neuron rate network with 4000 state variables, and estimate the largest Lyapunov exponent (LLE) with Benettin’s reshooting method. Two questions had to be answered before those numbers could be trusted:

1. **Is SRA1 at 400 Hz precise enough**, both without noise (where it can be compared with an adaptive reference) and with the noise the paper uses (where no adaptive reference exists)? And does the LLE depend on which integrator produced it?
2. **Do the two Lyapunov estimators in the code base agree** on the class the paper uses (`SRNNCellTypePairs`)? Benettin’s method and the QR full-spectrum method are independent algorithms; they had only been cross-checked on the older `SRNNModel2` class.

Both questions were asked in each of the preset’s three adaptation regimes, which span the dynamical range the paper is about: no adaptation (strongly chaotic, LLE around +3 to +4), single-timescale adaptation (weakly chaotic and intermittent, LLE around +0.3 to +0.9) and multiple-timescale adaptation (stable, LLE around -0.1 to -0.4).

2. Methods

2.1 Why trajectories cannot simply be compared

Two of the three regimes have a positive LLE. Any difference between two integrations, however small, grows as $\exp(\lambda t)$, so a pointwise comparison of whole trajectories measures the chaos, not the integrator. A 1e-10 adaptive reference is not exempt: it is a true trajectory of the system only locally, and after a few Lyapunov times it too is just one trajectory among many on the same attractor.

2.2 Reshooting: measuring integrator error without chaotic amplification

Precision is therefore measured the way Benettin measures the LLE. A reference trajectory is computed once. At each of many restart times the tested integrator is reset exactly to the reference state and integrated for one short segment; the distance from the reference at the segment’s end is the discretisation error accumulated over that segment, with no chaotic amplification because every segment starts from the truth. Two segment lengths are recorded: 0.02 s, which is Benettin’s renormalisation interval (so this is the error over one Benettin segment, the quantity that matters for the LLE), and two steps, the shortest span the fixed-step integrator accepts.

The error is the root-mean-square over state components, reported per state family (dendritic x, adaptation a, depression b) and relative to that family’s RMS value over the same window. Per condition and trial the reshoot uses up to 800 restarts for each segment length.

2.3 The references

Noise-free (sub-experiment A). Noise off. Reference: MATLAB ode45 with $\text{RelTol} = \text{AbsTol} = 1\text{e-}10$, output on a 1600 Hz grid, on the full 500-neuron network. Tested: SRA1 at 400, 800 and 1600 Hz. With the noise off SRA1’s drift stages reduce to Ralston’s second-order Runge-Kutta, so the segment error should fall 4x per halving of the step (slope 2 on a log-log plot). The preset’s piecewise activation has a derivative discontinuity at its breakpoints, which an adaptive solver steps around and a fixed-step scheme cannot; a measured slope between 1 and 2 would have indicated kink-dominated error.

With noise (sub-experiment B). Noise on at the preset’s $\text{sigma_u} = 0.025$. ode45 cannot integrate the SDE, so the reference is SRA1 itself at 12800 Hz, eight times the finest tested rate. The coarser runs consume the SAME Brownian path, rebuilt by exact aggregation (`coarsen_noise`): increments are summed, and the second stochastic integral (the area under the path within a step, which SRA1 uses) is carried with its base correction. Decimating the path would have produced a different path with the wrong variance. The difference between runs on one path is the strong error; SRA1 is strong order 1.5 for additive noise, so a slope of 1.5 is the theoretical floor, but where the drift error dominates the slope stays at 2. Eight-fold separation from the reference is the minimum the integrator test found necessary: closer, and the reference’s own error correlates with the tested run’s and flatters the slope.

LLE, integrator dependence (sub-experiment L). Noise off, 400 Hz, a 20 s run with Benettin accumulating over the last 10 s. Benettin with ode45 at 1e-10 for both the fiducial and the perturbed trajectory, versus Benettin

with SRA1 for both. Because Benettin uses one integrator for both trajectories most of the discretisation error is common to the two and cancels in their difference, so the exponents should agree more closely than the raw trajectory error suggests.

Benettin vs QR (sub-experiment C). The QR method integrates an $N \times N$ variational system per segment and is not feasible at 4000 states. It was run on a reduced network built from the same preset physics: $n = 40$, in-degree 8 (the preset’s connection density), `F_tracks_network = true` so the theoretical spectral radius matches the full network, noise off, ode45, $T = [-10, 10]$ s with 10 s of warm-up and 10 s of accumulation from $t = 0$. Both methods ran on the same fiducial trajectory. The largest QR exponent should equal Benettin’s LLE.

2.4 Trials

Every sub-experiment was repeated on five networks per condition, with `rng_seeds = [k, k+1]` for trial k , which draws the weight matrix, the stimulus amplitudes, the initial state and the per-neuron setpoints. Within a trial all sub-experiments share the network. The trial dimension exists because the earlier single-seed runs (2026-09-10) found that the 10 s finite-time LLE of the intermittent single-timescale regime scatters by about ± 0.2 from one trajectory to the next with either integrator, so a single seed cannot separate integrator bias from finite-time scatter. Paired per-trial values can: bias would show as a consistent sign across trials, scatter as differences of either sign.

2.5 A methodological trap that was found and closed

The stimulus is a three-step pattern (off / on / off). Its step edges fall at $T/3$ and $2T/3$ rounded to the nearest sample, and the linear interpolant ramps over one sample, so runs at different sampling rates see slightly different input around each edge. A first version of the analysis let the reshoot window straddle an edge, and one edge-crossing segment dominated the RMS: the error fell 3x from 400 to 800 Hz and then 23x to 1600 Hz, the one rate sharing the reference grid. All reshoot windows are confined to the stimulus-on middle third, where the input is identical on every grid.

2.6 Implementation

- `src/analysis/run_numerics_verification.m`: the stage; the four sub-experiments per trial and condition, with a cost table per run mode.
- `src/analysis/SRNNNumericsProbe.m`: a subclass of `SRNNCellTypePairs` that only widens access (inject a Brownian path, keep the path a run consumed, integrate a segment from a supplied state). It adds no physics.
- `src/model/integrators/coarsen_noise.m`: exact aggregation of a Brownian path (increment and second integral) onto a coarser grid.
- `src/figures/fig_numerics_verification.m`: the three figures.
- `scripts/paper/numerics_verification_trials_{config,run}.m`: this run.
- Tests: `test_numerics_probe` (injection, bit-identical segment re-integration from a mid-run state with noise, coarsening identities), `test_sde_integrators` (strong orders at large sigma), `test_run_modes`.

2.7 Settings of this run

setting	value
preset	<code>celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond</code>
network	$n = 500$, in-degree 100, 4000 state variables, piecewise activation, per-neuron setpoint
noise	<code>sigma_u = 0.025</code> (input-referred), additive on x
run mode	medium, 5 trials
A: noise-free reshoot	ode45 1e-10 at 1600 Hz; SRA1 at 400/800/1600 Hz; $T = 12$ s, window 4.3-7.99 s
B: noisy reshoot	SRA1 12800 Hz reference; SRA1 at 400/800/1600 Hz on the same path; $T = 4.5$ s, window 1.8-2.99 s
L: LLE	Benettin, 400 Hz, $T = 20$ s, accumulate 10-20 s, <code>lya_dt</code> 0.02 s, <code>d0 = 1e-3</code>

setting	value
C: Benettin vs QR	n = 40, in-degree 8, F_tracks_network, ode45 1e-9, T = [-10, 10] s, warm-up 10 s
restarts per segment length	up to 800

3. Acceptance criteria

Stated from theory and from the earlier single-seed runs (2026-09-10), before this ensemble was examined:

check	criterion	basis
noise-free convergence	fitted slope of segment error vs step within 1.8-2.2 on every seed and regime	SRA1's drift is second-order RK
noisy convergence	slope ≥ 1.5 on every seed and regime	strong order 1.5 is the floor
noise-free error at 400 Hz	relative error per 0.02 s segment $< 1e-4$ in every regime	the Benettin perturbation is $1e-3$ in norm; per-segment error must sit well below it
integrator dependence of the LLE	paired mean (SRA1 - ode45) within two standard errors of zero in every regime	no systematic bias
Benettin vs QR	paired median	difference

4. Results

Figures: `fig_numerics_solver/Fig_numerics_solver.png` (seed 1 traces, error per state family, convergence), `fig_numerics_lya_method/Fig_numerics_lya_method.png` (seed 1 local-exponent overlay and QR spectra), `fig_numerics_ensemble/Fig_numerics_ensemble.png` (paired per-seed comparisons). Data: `data/numerics_verification_trials/numerics_verification/numerics_verification_data.mat`.

4.1 Noise-free precision of SRA1 (sub-experiment A), five seeds per regime

RMS error per state component over one 0.02 s segment, SRA1 reset to the ode45 ($1e-10$) reference at every restart; mean over seeds, with the range.

regime	400 Hz	800 Hz (seed 1)	1600 Hz (seed 1)	slope, all seeds	state RMS (x)	relative at 400 Hz
no adaptation	3.8e-5 (1.8-5.7e-5)	8.8e-6	2.2e-6	2.01, 2.01, 2.01, 2.01, 2.01	3.6	1.1e-5
single-timescale	2.6e-6 (2.3-3.3e-6)	6.1e-7	1.5e-7	2.00, 2.01, 2.00, 2.01, 2.01	0.56	4.6e-6
multiple-timescale	7.3e-8 (4.5-8.7e-8)	2.2e-8	5.4e-9	2.00, 2.01, 2.01, 2.00, 2.00	0.29	2.5e-7

Per state family on seed 1 at 400 Hz (x / a / b): 3.5e-5 / - / - ; 4.8e-6 / 9.8e-7 / 3.5e-7 ; 2.2e-7 / 6.0e-8 / 2.5e-8. The dendritic state carries the largest error and depression the smallest, in proportion to their time constants. The two-step (local) error is 1.0e-5, 6.5e-7 and 2.2e-8. **Criterion met** on all 15 combinations.

4.2 With noise, shared Brownian path (sub-experiment B), five seeds per regime

regime	400 Hz, mean (range)	800 Hz (seed 1)	1600 Hz (seed 1)	slope, all seeds
no adaptation	4.1e-5 (2.9-5.3e-5)	1.2e-5	3.0e-6	2.00, 1.99, 1.98, 1.99, 2.00
single-timescale	1.2e-5 (1.1-1.3e-5)	3.0e-6	8.0e-7	1.95, 1.97, 1.96, 1.92, 1.95
multiple-timescale	4.8e-6 (4.5-5.0e-6)	1.2e-6	3.4e-7	1.87, 1.88, 1.86, 1.84, 1.82

The noisy error at 400 Hz is 1.1x, 4.6x and 66x the noise-free error in the three regimes: where the drift error is large (chaotic) the noise adds nothing measurable, and where it is tiny (stable) the stochastic term shows through, pulling the slope from 2 toward the strong-order floor of 1.5 without reaching it. **Criterion met** on all 15 combinations; the noisy slopes are remarkably consistent across seeds (within 0.06).

4.3 Integrator dependence of the LLE (sub-experiment L), 25 seeds per regime

Benettin over the last 10 s of a 20 s run at 400 Hz, noise off. Each seed is one network; both integrators start from its initial state.

regime	ode45 1e-10, mean +- sd (range)	SRA1 400 Hz, mean +- sd (range)	paired SRA1 - ode45, mean +- sd	s.e.m.	signs +/-	corr
no adaptation	+3.62 +- 0.62 (2.16 to 4.82)	+3.59 +- 0.72 (1.77 to 4.62)	-0.033 +- 0.448	0.090	14 / 11	0.79
single-timescale	+0.58 +- 0.17 (0.32 to 0.96)	+0.63 +- 0.18 (0.33 to 1.03)	+0.046 +- 0.170	0.034	15 / 10	0.54
multiple-timescale	-0.124 +- 0.016 (-0.146 to -0.070)	-0.124 +- 0.016 (-0.146 to -0.071)	-0.0000 +- 0.0000	0.0000	15 / 10	1.000

The paired means are 0.4, 1.4 and 0 standard errors from zero. **Criterion met** in all three regimes. Two further observations:

- In the stable regime the two integrators agree on every seed to the fourth decimal. A decaying perturbation aligns with the slowest direction and the finite-time exponent is then a property of the network, not of the trajectory.
- In the two chaotic regimes the paired scatter (sd 0.45 and 0.17) is of the same size as the seed-to-seed scatter (sd 0.62 and 0.17). After a few Lyapunov times the two integrators are on different trajectories of the same attractor, so a 10 s finite-time exponent from each is an independent draw. The 2026-09-10 controls established that changing the initial condition by 1% produces the same scatter with a single integrator, and that refining SRA1 to 1600 Hz does not move its value toward ode45's. The correlation of 0.79 in the chaotic regime shows that the network identity still explains most of the variance there.

4.4 Benettin vs QR on the reduced network (sub-experiment C), 25 seeds per regime

n = 40, in-degree 8, spectral radius matched to the full network, noise off, ode45, 10 s of accumulation after 10 s of warm-up.

regime	Benettin, mean +- sd (range)	QR lambda_1, mean +- sd (range)	paired QR - Benettin, mean +- sd	median abs	max abs	corr
no adaptation	-5.46 +- 4.42 (-10.0 to +1.99)	-5.45 +- 4.40 (-10.0 to +1.99)	+0.010 +- 0.065	0.005	0.25	0.9999
single- timescale	-0.17 +- 0.31 (-0.65 to +0.60)	-0.15 +- 0.29 (-0.50 to +0.64)	+0.015 +- 0.058	0.031	0.15	0.985
multiple- timescale	-0.117 +- 0.020 (-0.200 to -0.098)	-0.117 +- 0.004 (-0.129 to -0.110)	+0.0001 +- 0.020	0.005	0.08	0.02

Criterion met in all three regimes. What the ensemble adds to the single-seed picture:

- The 40-neuron no-adaptation network is heterogeneous across seeds: some seeds are dead fixed points (LLE $-10 = -1/\tau_d$), some sit at the edge of chaos, two are weakly chaotic ($+1.0, +2.0$). Benettin and QR track each other across that whole range; the largest disagreement (0.25, seed 22) is on a seed with LLE -6.5 , i.e. 4%.
- In the single-timescale regime several seeds land exactly on -0.500 , which is $-1/(2 \text{ s})$, the recovery rate of the single depression timescale: those seeds have settled to a fixed point whose slowest direction is depression recovery. The others are bursting or weakly chaotic, and the two methods agree to a median of 0.03.
- In the multiple-timescale regime QR's λ_1 is -0.117 ± 0.004 on every seed, which is the slowest adaptation recovery ($\tau_a = 10 \text{ s}$ gives -0.100 ; with the offset and depression it lands at -0.117). Benettin gives the same median but scatters ± 0.02 with two low outliers ($-0.153, -0.200$). In a strongly stable system the Benettin perturbation decays along whichever direction it started in until the slowest one dominates, and 10 s of warm-up is not always enough for that alignment; the finite-time estimate is then biased *negative*. The correlation of 0.02 reflects that QR's values have no spread to correlate with. This is the one systematic method effect in the study, and it can only make a stable network look more stable, never an unstable one look stable. The full-network stable-regime LLE of -0.12 in 4.3 is the same number, so the 500-neuron and 40-neuron networks agree on what sets stability there.

5. Interpretation

5.1 The integrator

SRA1 at 400 Hz is second-order accurate on this network in every regime, on every seed, with the noise on or off, and the piecewise activation's breakpoints do not degrade the order. The per-segment relative error runs from $1\text{e-}5$ in the chaotic regime to $3\text{e-}7$ in the stable one. What that error means for a trajectory depends on the regime:

- In the contracting regimes it does not accumulate: it saturates at about the per-segment error times the number of segments in one relaxation time. At an LLE of -0.12 that is about 400 segments, so an SRA1 trajectory tracks the true one to a few $1\text{e-}5$ per state indefinitely.
- In the chaotic regimes no trajectory can be tracked by any integrator for more than a few Lyapunov times: a relative error of $1\text{e-}5$ growing at 3.6 per second reaches order one in about 3 s, and a $1\text{e-}10$ reference buys about three more. Row 1 of the solver figure shows exactly that. The paper's stability claims must therefore rest on statistics, and the statistic it uses is the LLE.

5.2 The Lyapunov exponent

The LLE is integrator-independent in the ensemble mean, in every regime, to within its standard error over 25 seeds. What the ensemble makes unmistakable is the size of the finite-time scatter in the chaotic regimes: sd 0.6 in the strongly chaotic regime and 0.17 in the intermittent one over a 10 s window, with either integrator, and with the paired difference between integrators as large as the seed-to-seed difference. Two consequences:

- A single 10 s Benettin run in either chaotic regime is one draw from a distribution with that width. The

number to quote is the median over reps, which the sweeps compute; a single-realisation exponent should be read with ± 0.6 or ± 0.2 in mind.

- The stable regime has essentially no finite-time scatter (sd 0.016 across seeds, 0.0000 between integrators). Its exponent is set by the slowest adaptation recovery, and the same value emerges on 500 and on 40 neurons.

5.3 Benettin vs QR

The two independent estimators agree on `SRNNCellTypePairs` to a median of 0.005-0.03 across exponents from -10 to +2, and the residual disagreement in the stable regime is a known finite-time property of Benettin’s method (incomplete alignment within the warm-up) that biases toward more negative values. This corroborates the class’s Benettin implementation and its analytic-Jacobian QR implementation against each other, as was already established on `SRNNModel2`.

5.4 Interpretation of the ensemble figure

Row 1 of `Fig_numerics_ensemble`: points scatter about the identity line with no offset in the chaotic regimes and lie on it in the stable one. Row 2: points lie on the identity line across three orders of magnitude in the no-adaptation column; the stable column’s horizontal band is QR’s tight -0.117 against Benettin’s scatter, as discussed. Row 3: the reshoot error varies by a factor of three across seeds in the chaotic regime (the reference trajectories differ) and by 30% or less in the adapted regimes, while the slopes do not vary at all.

6. Remaining concerns

1. **Benettin in strongly stable regimes is biased negative when the warm-up is shorter than the alignment time.** Two of 25 stable-regime seeds on the reduced network show it (by 0.04 and 0.08). The full network’s stable regime showed no such scatter across integrators, but its alignment time has not been measured. Where a stable-regime LLE is quoted to better than 0.05, either lengthen the warm-up or report QR on a reduced network alongside.
2. **The finite-time scatter was measured noise-free.** With the paper’s noise on, the LLE is a property of the noisy dynamics and the trajectory-to-trajectory scatter could be narrower or wider. The reps spread at fixed parameters in the existing sensitivity sweeps is that measurement; it should be read off before the manuscript states a scatter.
3. **The reduced network is not the paper’s network.** Forty neurons with in-degree eight reproduce the three regimes qualitatively and show that the two Lyapunov methods agree on them, but individual reduced networks range from dead fixed points to weak chaos under the “no adaptation” label. The 500-neuron exponents come from Benettin only.
4. **The stochastic increment handling is verified only where it is visible.** At `sigma_u = 0.025` the drift error dominates, so the noisy slopes near 2 verify the drift on this network and only glimpse the strong-order term. The strong order itself (1.73 measured) is verified separately at large sigma in `test_sde_integrators`.
5. **Seed means network only when the generator is pinned.** This stage pins `twister` in every worker job; the rest of the pipeline does not, so the sweeps’ “seed k” (built on workers, `threefry`) is a different network from a figure’s “seed k” (built on the client). Recorded as a known bug in `UserNotes.md`; statistically harmless, but it forbids claiming that a figure shows one of the sweep’s reps.
6. **The analytic Jacobian’s finite-difference check is not preserved here.** It exists in `test_SRNNCellTypePairs` as a console result; priority 2 of the ranked list asks for it to be saved with its configuration and tolerance.

7. Reproduction

```
setup_paths();
wait_for_parpool(13);
numerics_verification_trials_run
```

% holds a PCT licence seat; polls if none is free
% ~50 min on 13 workers, then three figures

Earlier single-seed runs and the initial-condition / `lya_dt` / window controls:

- `figs/numerics_verification_test/interpretation_report_fast.md`
- `figs/numerics_verification_test_med/interpretation_report_medium.md`
- `scripts/examples/lle_finite_time_variability.m`